OSTEOPOROSIS DETECTION

Using Deep Learning Techniques

PROBLEM DOMAIN & **MOTIVATION**

- Osteoporosis is a progressive and asymptomatic bone disease that weakens bones and increases fracture risk.
- It often remains undetected until fractures occur, leading to disability and reduced quality of life.
- DEXA scans, the standard diagnostic method, are costly, time-consuming, and limited in availability.
- Knee X-rays are routinely available in hospitals but are not effectively used for osteoporosis screening.
- Manual interpretation of X-rays is subjective and error-prone.
- There is a strong need for an automated, low-cost, and accurate screening approach.
- Deep learning enables learning complex bone patterns from medical images.

- Osteoporosis incidence is rapidly increasing due to aging populations, especially in developing countries.
- Many outpatient settings lack access to specialized bone density equipment.
- Early-stage conditions like osteopenia are difficult to identify using conventional visual inspection.
- Delayed diagnosis results in higher treatment costs and irreversible bone damage.
- Machine learning models can assist clinicians as decision-support tools, not replacements.
- Automated screening can help in large-scale population health monitoring.
- Lack of explainability in AI systems reduces clinical trust and adoption.

EXISTING SYSTEM— ANALYSIS

- DEXA Scan is the current gold standard for osteoporosis diagnosis.
- It measures bone mineral density (BMD) but requires specialized equipment.
- DEXA machines are expensive and not portable, limiting accessibility.
- Diagnosis often occurs only after significant bone loss.
- Traditional radiographic analysis depends on manual interpretation by experts.
- Manual assessment is time-consuming, subjective, and prone to inter-observer variability.
- Conventional computer-aided systems rely on handcrafted features, limiting accuracy.
- Most existing ML models focus only on binary classification (normal vs osteoporosis).

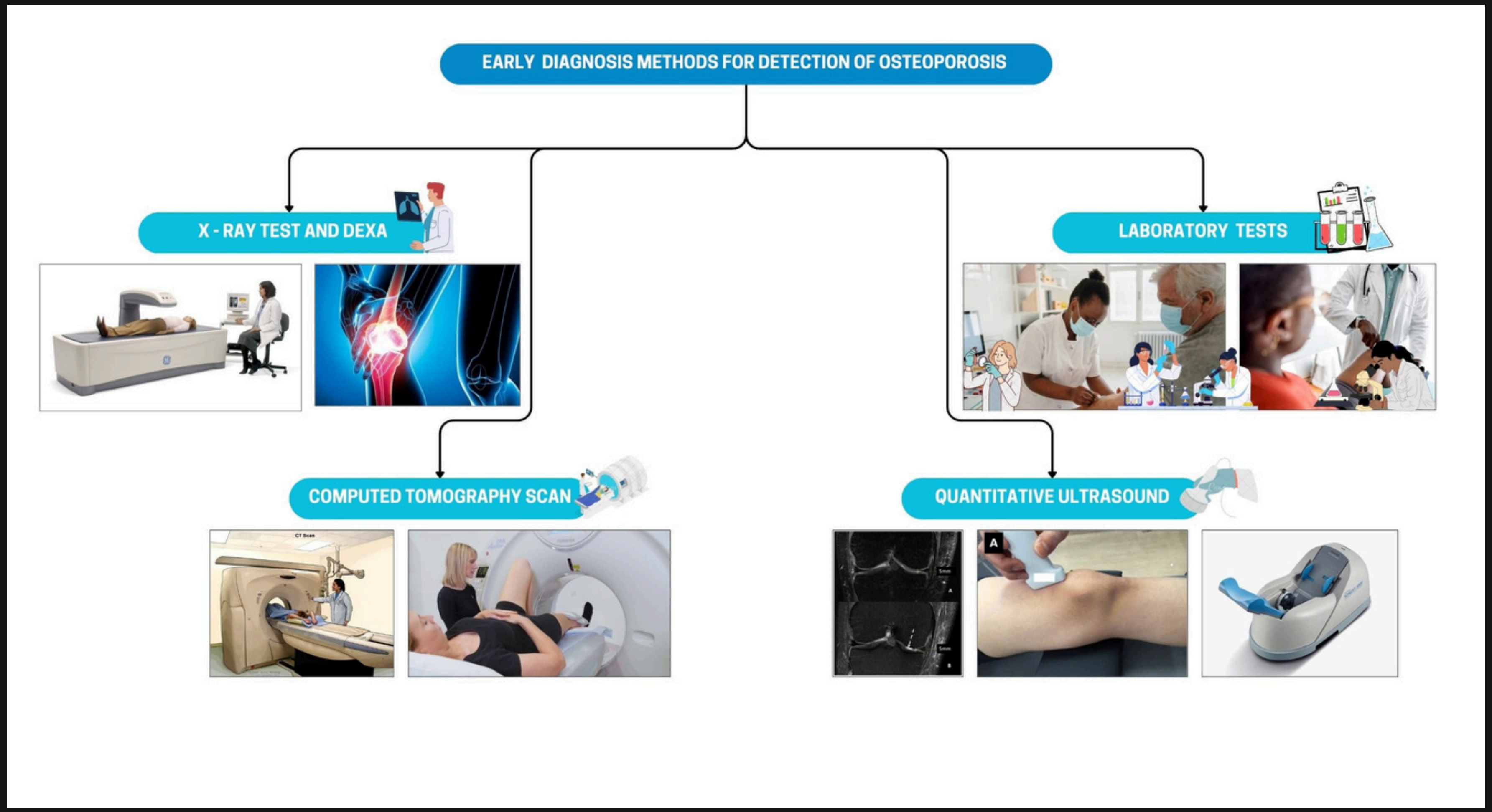
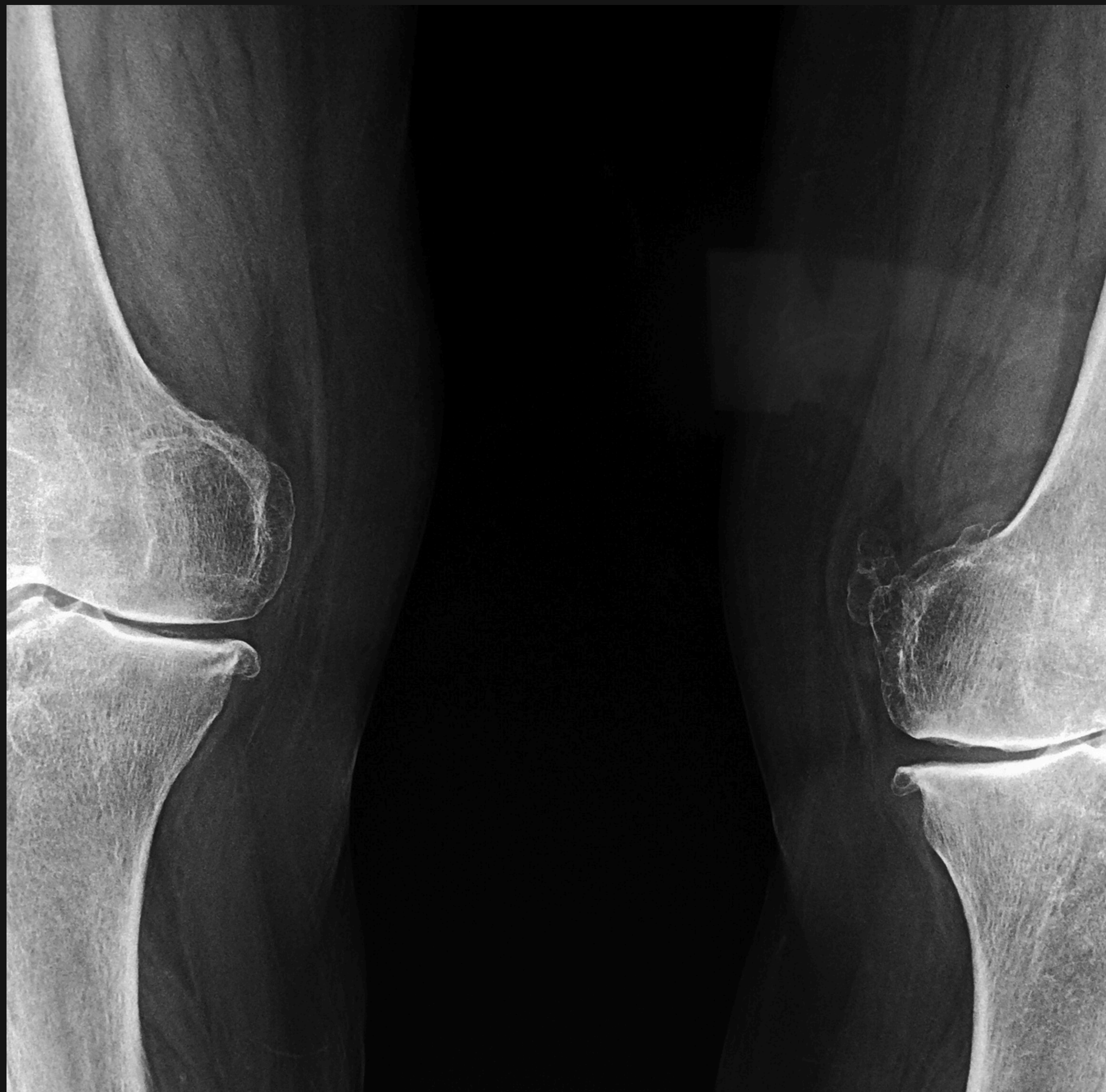


Figure 1: *Existing Methodologies*



FEASIBILITY STUDY AND LIMITATIONS OF EXISTING SYSTEMS

LIMITATIONS OF EXISTING SYSTEMS

- High dependency on DEXA equipment, which is costly and infrastructure-intensive.
- Limited availability of DEXA scanners in rural and semi-urban regions.
- Late diagnosis due to absence of routine screening mechanisms.
- Manual X-ray analysis suffers from subjectivity and human error.
- Traditional CAD systems show poor generalization on unseen datasets.

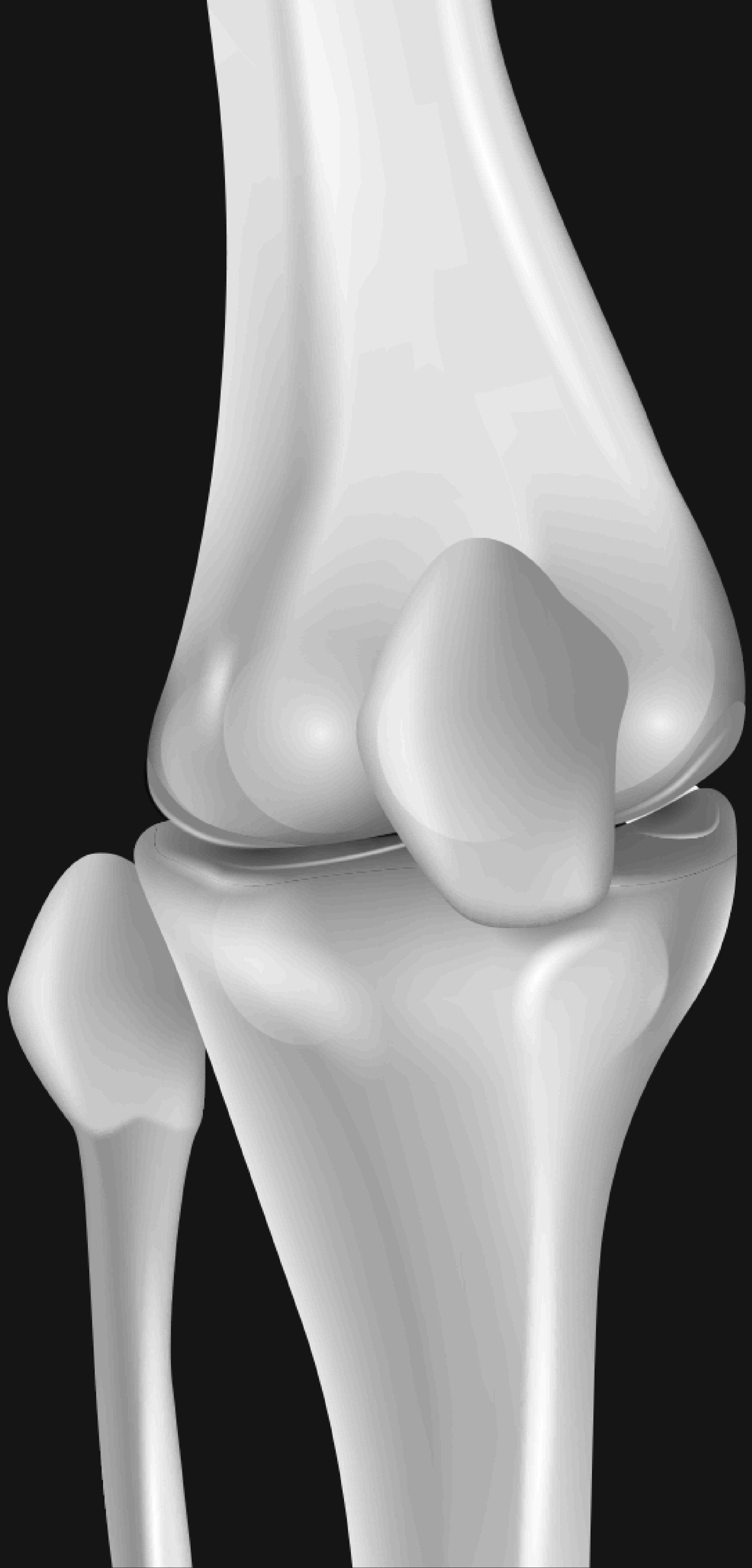
FEASIBILITY OF PROPOSED SYSTEM

- Knee X-ray images are already available in routine clinical practice.
- Transfer learning enables high performance with limited medical data.
- GPU-based training makes the system computationally feasible.
- Deployment through Streamlit allows real-time and user-friendly access.
- Explainable AI improves clinical acceptance and trust.

"The proposed system is economically, technically, and operationally feasible using existing radiology infrastructure."

DATASET

Dataset Source	Normal	Osteopenia	Osteoporosis	Type
Dataset 1 – Kaggle (Binary)	372	–	372	Binary
Dataset 2 – Kaggle (Binary)	186	–	186	Binary
Dataset 3 – Kaggle (Binary)	186	–	186	Binary
Dataset 4 – Mendeley (Multiclass)	36	154	Included (count not specified)	Multiclass
Dataset 5 – Using for this project	780	374	793	Multiclass



OBJECTIVES

- To develop a deep learning–based system for automated osteoporosis detection using knee X-ray images.
- To apply transfer learning techniques for improving classification accuracy on limited medical datasets.
- To perform multi-class classification into normal, osteopenia, and osteoporosis categories.
- To address class imbalance issues using proper data handling techniques.
- To integrate explainable AI methods such as SHAP or Grad-CAM to interpret model decisions.
- To evaluate the model using clinical performance metrics like precision, recall, F1-score, and ROC-AUC.
- To design a user-friendly interface for real-time screening support.

These objectives ensure not only high accuracy but also fairness, interpretability, and clinical usability of the system

METHODOLOGY

This pipeline ensures accurate, interpretable, and clinically meaningful osteoporosis detection from knee X-rays

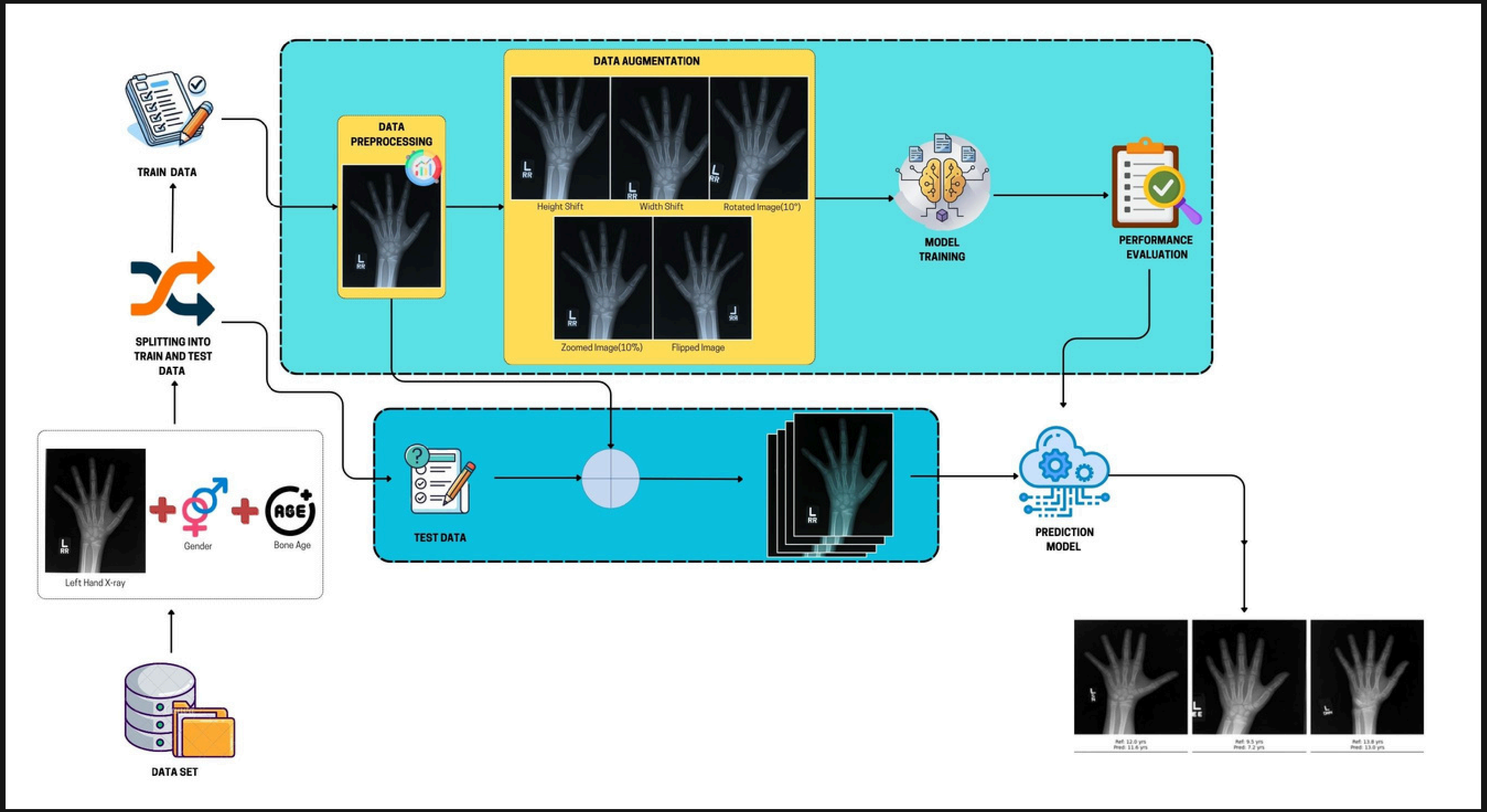


Figure 2: *Proposed Methodology*

THANK YOU